

GPS Time Sync Engineering Guide

Source-level architecture and maintenance notes for GPS Time Sync 1.20

Version 1.20 | Windows desktop utility | VK-172 USB GPS receiver

Last changed: July 17, 2026

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Scope

This guide describes the current Delphi/VCL implementation using GPSTimeSync.pas, GPSTimeSync.dfm, and GPSTimeSync.dpr. It is intended for maintainers who need to understand the application structure, serial-port workflow, GPS parsing, clock synchronization, tray behavior, Help integration, and deployment files.

Project Structure

File or folder	Role
GPSTimeSync.dpr	Program entry point, single-instance lock, first-run behavior, and form creation.
GPSTimeSync.pas	Main form logic, serial I/O, WMI detection, NMEA parsing, timers, logging, time sync, tray behavior, and Help button handler.
GPSTimeSync.dfm	IDE-editable VCL form layout and component definitions.
Help\index.html	HTML Help entry point opened from the application Help button.
source	GitHub Pages website source.
source\downloads	Website download payloads, including installer, EULA, and public guide PDFs.
docs	Editable Word guide sources.

Runtime Startup

1. GPSTimeSync.dpr creates or opens GPSTimeSync.lock with exclusive sharing to prevent multiple running instances.
2. If the lock cannot be acquired, the application shows 'Another instance of this application is running.' and exits.
3. The program checks for gps_settings.ini beside the executable. If it is absent, it shows the main form for parameter editing.
4. The main form is created from GPSTimeSync.dfm and initialized through TfrmGPSTimeSync.FormCreate.

Implementation note. The code checks for gps_settings.ini and declares IniFilePath, but the current GPSTimeSync.pas source does not implement INI persistence.

Main Form Components

Component group	Purpose
Position/status fields	txtLatLong, edtAltitude, edtFixQuality, edtFixMode, edtSatInUse, and edtSatInView display parsed GPS state.
Time display	txtDate and lblSystemTime show GPS-derived local or UTC time when available; otherwise they show current Windows time.
Sync settings	edtComPort, edtSyncInterval, edtDriftSeconds,

Component group	Purpose
	rdoSetUTC, rdoSetLocal, and edtEnableLogging control runtime behavior.
Action buttons	Hide To Tray, Save, Close, and Help provide primary user actions.
Timers	LEDFlashTimer polls serial data, tmClock updates display/LED state, and tmTimeCheck performs drift comparison.
TrayIcon1	Keeps the application accessible while hidden.

Serial Port and VK-172 Detection

FindVK172GPSDongle uses WMI against Win32_SerialPort and searches for PNPDeviceID containing VID_1546&PID_01A7. If found, the matching DeviceID such as COM3 is selected. Otherwise the user is prompted to choose a COM port manually.

Routine	Responsibility
PopulateComPorts	Loads COM1 through COM30 into the combo box and keeps any existing custom entries.
OpenSerial	Opens \\.\COMx, applies baud rate, 8 data bits, no parity, one stop bit, raw mode, and communication timeouts.
CloseSerial	Closes the handle if open.
ReadSerialAvailable	Uses ClearCommError/COMSTAT to detect queued bytes and reads up to 4096 bytes.
ComPortChanged	Reopens the port when the selected COM port changes.

NMEA Processing

Incoming serial chunks are accumulated until newline-delimited NMEA sentences are available. Valid sentences must begin with '\$', contain '*', and pass checksum validation.

Sentence	Handled fields
RMC	UTC date/time, valid/invalid status, latitude, and longitude.
GGA	Fix quality, satellites in use, and altitude converted from meters to feet.
GSV	Satellites in view.

Clock Synchronization

tmTimeCheckTimer compares Windows UTC, computed from Now through LocalToUTC DT, with the last valid GPS UTC. If the absolute difference is greater than or equal to the drift threshold, SyncSystemTime attempts to update the clock.

Mode	API path
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Mode	API path
Set Local	GPS UTC is converted to local time with <code>SystemTimeToTzSpecificLocalTime</code> , then <code>SetLocalTime</code> is called.
Set UTC	GPS UTC is passed to <code>SetSystemTime</code> .
Failure	The log records an administrator-rights error when Windows rejects the clock change.

Logging and LED Behavior

- `AddToLog` writes timestamped application messages to `GPSTimeChange.log`.
- `LogLine` appends incoming NMEA sentences to the visible memo when `Enable Logging` is checked.
- The memo is capped at 500 lines.
- A valid NMEA sentence turns the LED green immediately.
- `tmClockTimer` updates the display once per second and resets the LED to red while waiting for the next valid sentence.

Help Button Integration

The Help button is an IDE-editable `TBitBtn` named `btnHelp` in `GPSControlPanel.dfm`. Its `btnHelpClick` handler builds a portable `Help\index.html` path beside the running executable, verifies that the file exists, and opens it with `ShellExecute`.

Known Maintenance Considerations

Area	Consideration
Settings persistence	Save currently applies settings for the running form but does not write <code>gps_settings.ini</code> .
Receiver compatibility	Automatic detection is specific to VK-172 hardware ID <code>VID_1546&PID_01A7</code> ; other receivers require manual COM selection.
Administrator privileges	Clock correction requires elevated rights.
Unicode and encoding	Keep source and HTML files in encodings that preserve degree symbols and comparison signs used in UI labels.
Help path	The current Help button target is a portable <code>Help\index.html</code> path under the application folder, so the installed package is not tied to a development-machine folder.
Single instance	<code>GPSTimeSync.lock</code> is closed on normal program exit in the DPR finally block.

Build and Release Notes

- Project file: `GPSTimeSync.dproj`.
- RAD Studio toolchain path from project directives: `C:\Program Files (x86)\Embarcadero\Studio\37.0\bin`.

- Website source: source\index.html and source\download.html.
- Website downloadable artifacts: source\downloads.
- After approved changes, review git status, stage only intended files, commit clearly, push to the Pages repository, and run the manual Pages workflow.